2201. Count Artifacts That Can Be Extracted

Solved

Medium Topics Hint

There is an $n \times n$ **0-indexed** grid with some artifacts buried in it. You are given the integer n and a **0-indexed** 2D integer array artifacts describing the positions of the rectangular artifacts where artifacts[i] = $[r1_i, c1_i, r2_i, c2_i]$ denotes that the ith artifact is buried in the subgrid where:

- $(r1_i, c1_i)$ is the coordinate of the **top-left** cell of the ith artifact and
- $(r2_i, c2_i)$ is the coordinate of the **bottom-right** cell of the ith artifact.

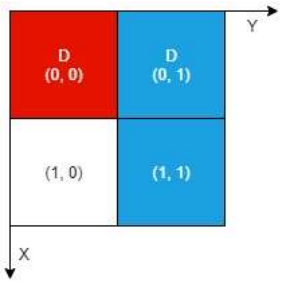
You will excavate some cells of the grid and remove all the mud from them. If the cell has a part of an artifact buried underneath, it will be uncovered. If all the parts of an artifact are uncovered, you can extract it.

Given a **0-indexed** 2D integer array dig where dig[i] = $[r_i, c_i]$ indicates that you will excavate the cell (r_i, c_i), return *the number of artifacts that you can extract.*

The test cases are generated such that:

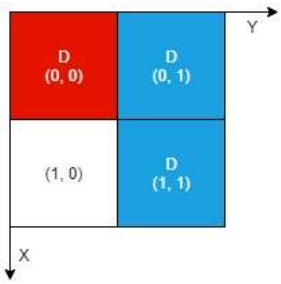
- No two artifacts overlap.
- Each artifact only covers at most 4 cells.
- The entries of dig are unique.

Example 1:



> **Input:** n = 2, artifacts = [[0,0,0,0],[0,1,1,1]], dig = [[0,0],[0,1]]
> **Output:** 1
> **Explanation:**
> The different colors represent different artifacts. Excavated cells are labeled with a 'D' in the grid.
> There is 1 artifact that can be extracted, namely the red artifact.
> The blue artifact has one part in cell (1,1) which remains uncovered, so we cannot extract it.
> Thus, we return 1.

Example 2:



> **Input:** n = 2, artifacts = [[0,0,0,0],[0,1,1,1]], dig = [[0,0],[0,1],[1,1]]
> **Output:** 2
> **Explanation:** Both the red and blue artifacts have all parts uncovered (labeled with a 'D') and can be extracted, so we return 2.

Constraints:

- $1 <= n <= 1000$
- $1 <= artifacts.length, dig.length <= \min(n^2, 10^5)$
- artifacts[i].length == 4
- dig[i].length == 2
- $0 <= r1_i, c1_i, r2_i, c2_i, r_i, c_i <= n - 1$
- $r1_i <= r2_i$
- $c1_i <= c2_i$
- No two artifacts will overlap.
- The number of cells covered by an artifact is **at most** 4.

- The entries of `dig` are unique.

Seen this question in a real interview before? 1/5

Yes No

Accepted 23,183 /40.6K | Acceptance Rate 57.1%

Topics

Hint 1

Hint 2

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